

Problem Statement - Part II

Question-1: What is the optimal value of alpha for ridge and lasso regression? What will be the changes in the model if you choose double the value of alpha for both ridge and lasso? What will be the most important predictor variables after the change is implemented?

Answer:

1. Optimal value of alpha for Ridge Regression = 10
2. Optimal value of alpha for Lasso = 0.001

Double the value of alpha for both ridge and lasso i.e., 20 and 0.002:

Metrics once we double the value for Ridge and Lasso are as below:

	Ridge Regression	Lasso Regression
Metric		
R2 Score for 'Train' dataset	0.93	0.94
R2 Score for 'Test' dataset	0.93	0.90
RSS for 'Train' dataset	65.99	62.08
RSS for 'Test' dataset	32.54	44.31
MSE for 'Train' dataset	0.06	0.06
MSE for 'Test' dataset	0.07	0.10
RMSE for 'Train' dataset	0.25	0.25
RMSE for 'Test' dataset	0.27	0.32

Important variables for Ridge now are:

GrLivArea	1.24
OverallQual_9	1.22
OverallQual_8	1.20
Functional_Typ	1.19
Neighborhood_Crawfor	1.19
Exterior1st_BrkFace	1.17
OverallCond_9	1.16
TotalBsmtSF	1.15
OverallCond_7	1.13
Neighborhood_StoneBr	1.13
Name: Ridge, dtype: float64	

Important variables for Lasso now are:

OverallQual_9	1.62
GrLivArea	1.39
OverallQual_8	1.38
Neighborhood_Crawfor	1.33
OverallCond_9	1.27
Exterior1st_BrkFace	1.22
Functional_Typ	1.20
Neighborhood_Somerst	1.20
TotalBsmtSF	1.18
OverallQual_7	1.16
Name: Lasso, dtype: float64	

Question-2: You have determined the optimal value of lambda for ridge and lasso regression during the assignment. Now, which one will you choose to apply and why?

Answer: Use case at hand will decide what model we shall choose.

- If there are too many variables and our primary goal involves feature selection, then we will use Lasso.
- If we don't want large coefficients and reduction of coefficient magnitude is one of our prime goals, then we will use Ridge Regression.
- For current use-case, would choose Lasso as it also does feature selection

	Ridge Regression	Lasso Regression
Metric		
R2 Score for 'Train' dataset	0.94	0.95
R2 Score for 'Test' dataset	0.93	0.88
RSS for 'Train' dataset	59.25	48.94
RSS for 'Test' dataset	33.03	51.60
MSE for 'Train' dataset	0.06	0.05
MSE for 'Test' dataset	0.08	0.12
RMSE for 'Train' dataset	0.24	0.22
RMSE for 'Test' dataset	0.27	0.34

Question-3: After building the model, you realised that the five most important predictor variables in the lasso model are not available in the incoming data. You will now have to create another model excluding the five most important predictor variables. Which are the five most important predictor variables now?

Answer: Top 5 predictors now are:-

1. 2ndFlrSF
2. Exterior1st_BrkFace
3. 1stFlrSF
4. TotalBsmtSF
5. Functional_Typ

2ndFlrSF	0.30
Exterior1st_BrkFace	0.24
1stFlrSF	0.21
TotalBsmtSF	0.18
Functional_Typ	0.17

Name: Lasso, dtype: float64

Question-4: How can you make sure that a model is robust and generalisable? What are the implications of the same for the accuracy of the model and why?

Answer:

- Model should be simpler
- Robust model is one whose performance is not affected much even when there is variation in the data
- A generalizable model is able to adapt properly to new, previously unseen data, drawn from the same distribution as the one used to create the model.
- To make sure a model is robust and generalizable, we have to take care it doesn't overfit. This is because an overfitting model has very high variance and a smallest change in data affects the model prediction heavily. Such a model will identify all the patterns of a training data, but fail to pick up the patterns in unseen test data.
- In other words, the model should not be too complex in order to be robust and generalizable.
- If we look at it from the perspective of Accuracy, a too complex model will have a very high accuracy. So, to make our model more robust and generalizable, we will have to decrease variance which will lead to some bias. Addition of bias means that accuracy will decrease.
- In general, we have to find strike some balance between model accuracy and complexity. This can be achieved by Regularization techniques like Ridge Regression and Lasso.